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Successful implementation of telemedicine depends on personal relations between company representatives and healthcare providers: A qualitative study of business models for Danish home telemonitoring

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Abstract

The struggle to transform telemedicine from project level to daily clinical practice is ongoing. An assessment of the business models of telemedicine applications could provide insights into how to facilitate this transition. Our aim was to identify the qualitative characteristics of business models of long-term operational telemedicine providers in Denmark.

A structured interview study design was applied to interviews of representatives from seven Danish companies providing home telemonitoring in long-term operation. Data was analysed using Osterwalder's Business Model Canvas framework.

Multiple themes emphasized the importance of strong personal relationships between company representatives and healthcare providers. Personal relations could 1) secure a strong relationship to lead users and clinical ambassadors; 2) facilitate work with healthcare providers to develop, test and revise value propositions; 3) promote user support and education; 4) establish an indirect connection between companies and healthcare managers or decision makers.

Thus, a strong personal relationship between company representatives and healthcare providers is of paramount importance when integrating home telemonitoring from project stage into clinical practice. However, this strategy could lack patient involvement, use of data, and business scalability. Additionally, companies with the ability to establish strong personal connections could be favoured over companies which provide strong clinical and economic evidence.

Keywords

Business model, home telemonitoring, implementation, organization and administration, qualitative study, telemedicine

Introduction

Telemedicine is an increasingly important tool for providing affordable and high-quality healthcare services.[1–4] It encourages and enables patients to take responsibility for their own health and treatment,[2] improves outcomes,[5] and reduces the number of physically provided healthcare services.[6] Despite its proven validity in small scale projects, telemedicine struggles to transition from research to clinical practice.[7,8] Thus there is a need for research focused on the implementation of telemedicine.[9] A lack of integration into clinical practice is found within many healthcare services and has been suggested to arise at the provider level.[10,11] Providers of telemedicine are often paid to help develop applications through project funding, but once the project phase ends a sustainable business model is often missing, resulting in commercial failure.[6,8] Studies indicate that developing a comprehensive business model could enhance the likelihood of transition to operation.[12,13] Previous studies which have investigated business models for clinically integrated telemedicine used quantitative data and

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focused on long-term operational applications.[8,14] In contrast, acknowledged models for the proper evaluation and implementation of healthcare services have included both quantitative and qualitative measures.[11,15,16] The qualitative aspects of business models for long-term operational telemedicine are still poorly understood. In addition, the organisation and financing of healthcare services differ between countries, which challenges the generalisation of telemedicine business models. Thus, there is a need to investigate telemedicine business models in countries with a uniform healthcare reimbursement system. Denmark has a public tax-paid healthcare system, which fulfils this condition.[6] We hypothesised that a qualitative analysis of selected long-term operational telemedicine services in Denmark could provide key insights into how successful telemedicine is accomplished in clinical practice. Using Osterwalder's Business Model Canvas we aimed to identify the qualitative characteristics of business models of long-term operational telemedicine providers in Denmark.

Methods

Our qualitative study used a thematic analysis of structured interviews with company representatives, investigating the business models (BMs) of seven providers of long-term operational telemedicine used in Denmark. Analysis of BMs should assess value creation, value delivery and value capture.[6] Osterwalder's Business Model Canvas (BMC) provides a framework for such analyses, dividing business models into nine elements, answering the following questions:[17]

1. Customer segments: For whom is value created?
2. Value propositions: What value is created?
3. Channels: How do value propositions reach customer segments?
4. Customer relationships: Which relations exist with each customer segment?
5. Revenue streams: How is income generated?
6. Key resources: Which resources are needed for the business model to work?
7. Key activities: Which activities are needed for the business model to work?
8. Key partners: Which partnerships are needed for the business model to work?
9. Cost structures: Which costs are related to the business model?

The framework is well-established and has been used in previous studies of telemedicine business models.[8,18,19] Thus, BMC was used to analyse constructs of BMs to ensure a joint understanding of and point of reference for describing how telemedicine providers create, deliver and capture value.

Sample

In total 418 Danish telemedicine initiatives were identified in February 2017 using 'The Danish Telemedicine Map'[20] as the data source. The scope of the present study was limited to the subgroup of telemedicine referred to as home telemonitoring (HT) since comparable data exists for this subgroup.[21] The following definition was adopted: *'Home telemonitoring includes telemedicine technologies through which data is transferred from the home of the patient to healthcare professionals in a digital fashion. Data is used in the clinical decision making by the healthcare professionals and includes both subjective patient-reported outcomes as well as quantitative data from measuring devices. Transfer of data is done either automatically or is manually typed. The definition does not include transfer solely by audio or video. Applications that allow the patient to leave home are included if patients are not admitted to the hospital'*.[21] Appendix 1 provides definitions used throughout this paper.

As seen in Figure 1, 306 initiatives were excluded since they did not involve HT or were ongoing projects. Contact persons of the remaining 112 initiatives were surveyed by phone or e-mail with the aim to identify HT initiatives that were in operation. The duration of the initiatives was registered along with names and contacts of the companies that were the primary providers of the HT applications (response rate 82%). Twenty-two initiatives were included and sorted by time in operation. Time in operation ranged from 2 months to 8 years and 1 month. The eight companies delivering HT applications which had been in operation the longest were asked to participate. Seven accepted the invitation (Figure 1). Of these companies, time in operation ranged from 3 years and 1 month to 8 years and 1 month (Table 1).

Participant characteristics

We conducted one interview per company with a varying number of participants present since the companies were free to decide which employee(s) were best suited to answer the interview questions. We conducted four one-to-one interviews, two 2-person interviews, and a focus group interview with 8 participants. Interviews ranged from 90–120 minutes. Table 2 lists the characteristics of the interview respondents.

Data collection and analysis

A structured interview guide was prepared and tested in two pilot interviews guided by the method described by

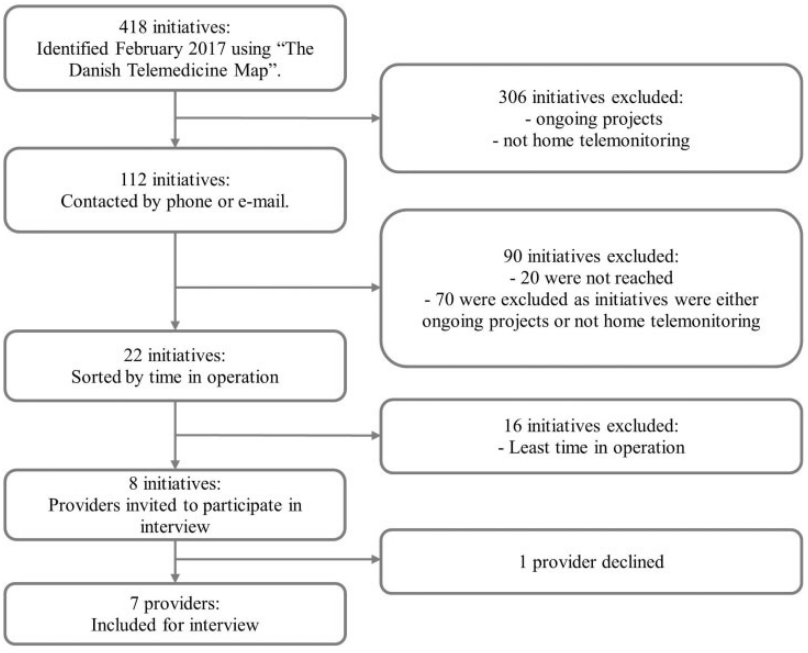

Figure 1. The inclusion of home telemonitoring providers.

Kvale and Brinkmann.[23] The aim was to articulate the essence of each BMC element in a single open-ended question using wording that would not require any prerequisite knowledge of BMC[17] (Appendix 2). Prior to the interviews, the respondents were briefed on the aim and procedure, asked to read the interview questions and watch a video to get acquainted with the BMC framework.[24] This was done to ensure a shared and standardised understanding of the framework. During the interviews, the interviewer consolidated and interpreted answers, wrote essential statements on Post-It® notes, and placed them accordingly on a poster displaying the nine elements of BMC. Respondents were encouraged to challenge the statements. This process provided continuous validation of interpretations during the interviews.[23] All interviews were conducted face-to-face in respondents' own facilities if possible. Two interviews were done using Skype and an online version of the BMC poster. All interview participants were informed about the purpose of the study, what was expected of them as a participant, and steps were taken to ensure confidentiality. Consent was obtained from all participants.

All interviews were transcribed and coded using a priori codes[23] based on the BMC framework[17] (Appendix 3). All codes were retrieved using QDA Miner Lite[25] and corresponding interview transcripts were read while elaborating the statements on the BMC poster from the interviews. This was done to consolidate meaning.[23] All updated BMC posters were sent to and validated by each respective respondent. Common traits within each element of the BMC were identified by sorting statements on the seven BMC posters according to which of the nine elements they originated from. For each element, statements were read and grouped into themes. Each theme was given a title and interpreted, and the number of companies (diversity) supporting the theme (Figure 2) was also determined.

To ensure the validity of this process, the grouping of statements was based on a consensus between the authors, and interview transcripts were reread if needed.[26] Because the aim was to identify an overarching trait among all BMs, themes were sorted by diversity, and the top three themes of each element of the BMC were chosen for further analysis. Less prominent but surprising or contradictory themes were also included for analysis. Quotes were included to illustrate essential points using Kvale and Brinkmann's guidelines for interview quotes.[23]

Results

The top three themes of each of the nine BMC elements sorted by number of companies supporting the theme (div) are presented in Table 3. Findings for each element of the BMC are elaborated in the following section.

Customer segments

Dominant themes within customer segments were *Patients, Payers,* and the *Secondary healthcare sector*. These themes supported a BM characterised by multi-sided markets as customer segments were

Table 1. Characteristics of included companies sorted by time in operation in descending order.

Company #	Medical specialty	Products/services other than home telemonitoring	Ownership	Size and market
1	Dermatology	Yes	Private	SME, national
2	Multiple specialties	No	Private	SME, multinational
3	Neurology	No	Private	SME, national
4	Cardiology	No	Private	SME, multinational
5	Multiple specialties	No	Public	SME, national
6	Cardiology	Yes	Private	Large, multinational.
7	Geriatric	No	Private	SME, national

SME: small- or medium-sized enterprise.[22]

Table 2. Characteristics of interview respondents.

Characteristics	Number of respondents (n = 16)
Female	10
Male	6
Education	
IT and engineering	4
Health	9
Business	3
Role	
IT and computer science	3
Research assistant	1
Product manager/business developer	4
Product specialist	6
CEO	2

interdependent. Patients or healthcare providers acted only as end-users and were not directly involved in payment for the applications. Some respondents articulated that even though they had someone pay for the HT application, they struggled to pinpoint the person making the decision to pay for the application:

> *'My primary contact is the healthcare providers, but of course they have a manager granting the money. But I just send the invoice and receive the money, I rarely have anything to do with the payers.'*

Other less supported customer segments were the *Primary healthcare sector* (div 4), *Researchers* (div 3), and *Society* (div 2).

Value propositions

Value propositions were *Better use of resources resulting in reduced time of treatment, Ease of use, and Employee satisfaction*. Value propositions were targeted mainly at healthcare providers working in hospitals and outpatient clinics. The applications helped them do their jobs better, through facilitating better use of resources. In addition, characteristic attributes were a high degree of usability and employee satisfaction. The most widely addressed patient value propositions were *Prevention of disease* (div 4) and *Improved quality of life* (div 4). *Clinical evidence* (div 3) and economic incentives such as *Cost savings* (div 3) were addressed by less than half of the companies.

Channels

Prominent channels were *End-users influencing payers, Conferences and fora, and Word of mouth*. All three targeted healthcare professionals. In answering why one company focused on always responding to phone calls and helping users, the respondent explained:

'We do that because, if the users are happy then they will let their management know that they want us. The users and especially the doctors can influence the hospital management to buy the application. For us it's a way of holding on to the hospital management'.

In addition, healthcare providers benefitted from all of the top three themes of value propositions. Thus, a focus on creating value for and satisfying the healthcare providers tapped into the two partner channels over which the companies have no direct control: *End-users influencing payers* and *Word of mouth*. *Conferences and fora* on the other hand represented an owned channel in which the companies could interact directly with payers and healthcare providers. Four companies articulated that patients were addressed through clinicians (div 4). Thus, a common trait among the three channels was personal interactions or relations between individuals, and partner channels were the most prominent.

Customer relations

The key themes under customer relations were *Personal assistance, Keeping customers,* and *Cocreation*. Thus, customer relations were highly based on personal relationships. No themes regarding self-service functions or

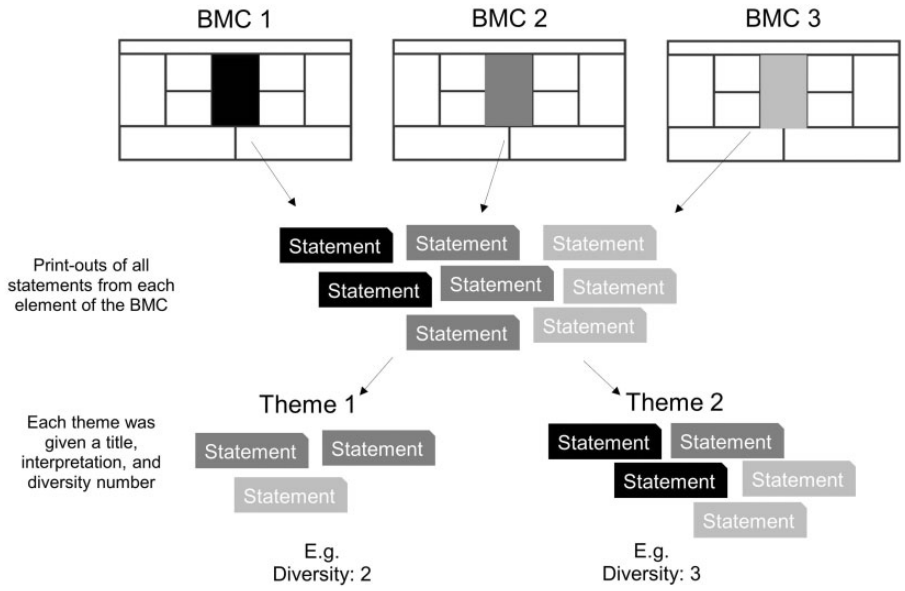

Figure 2. Meaning analysis and interpretation of interviews. For each element of the Business Model Canvas (BMC), interview statements were grouped according to themes. Each theme was given a title and interpreted, and the number of companies (diversity) supporting the theme was also determined.

automated services emerged. One respondent expressed that personal visits were inevitable:

> *'I think most of our clients get a visit at least once a year. When I look at it from a business perspective, I feel like I want to move away from personal visits, but you can't really ignore how well they work'.*

Personal visits were the most prominent way of interacting with users throughout all the companies investigated. These visits served multiple purposes, influencing other elements of the business model. They 1) Secured a strong relationship to *Lead users and clinical ambassadors*, thereby establishing relations to a key partner; 2) Facilitated *Cocreation by* developing, testing and revising value propositions in cooperation with healthcare providers. This was a way to support customer relations. During these sessions both lead users and end-users attended. End-users were mostly clinicians; 3) Promoted user *Support* and *Education*, thus playing an essential role in key activities; 4) Established an indirect connection between companies and healthcare managers or decision makers. Personal visits facilitated the critical task of keeping end-users satisfied, thereby maintaining an indirect channel to payers.

Key resources

Key resources were dominated by the need for human resources in the form of *Technical personnel* and *Healthcare personnel*. The most important type of physical resource was *IT infrastructure*. However, intellectual resources were inconspicuous, *Value of data* (div 4) and *Security* (div 3) being the most pronounced, and only one respondent mentioned *Patents* (div 1).

Key activities

Overall, key activities were *Support, Development, and Education*. These aligned with a strong need for human interactions to support customer relations, emphasising problem-solving activities like developing new solutions to individual customer problems. Companies hosted local sessions to teach end-users how to best make use of the HT applications. In addition, if issues arose during daily clinical work, most companies had an extensive support system handling technical issues that might occur on a day-to-day basis. Some companies organised these support systems by assigning direct company hotlines and dedicated support departments.

Key partners

Lead users and clinical ambassadors were stated to be key partners by all except one respondent. However, fewer companies supported the remaining top three partners *Local IT-departments* and *Other suppliers of healthcare IT*. The incentive for partners were acquisition of resources or activities, and a way of reducing risk and uncertainty. *Lead users and clinical ambassadors* helped facilitate local support of end-users and played a key role in helping companies develop HT applications since they possessed key insights into user issues and ideas for further development.

Table 3. Top three themes of each element of the business model canvas sorted by number of providers supporting a theme (diversity).

BMC element	Theme	Description	Diversity (div)
Customer segments	Patients	Patients were represented widely from elderly medical citizens and chroniclers to debut schizophrenics. They could be grouped in need of care, for example monitoring, rehabilitation, or treatment in the patient's own home.	7
	Payers	Payers could be anything from the individual municipality employees to decision-makers in the municipalities, individual departmental administrators at the hospital to governing units in the regions, including tendering groups.	7
	Secondary healthcare sector	Described users of the home telemonitoring (HT) application within the secondary healthcare sector. The users were represented within both psychiatry and the somatic field, the later worked in hospital wards and outpatient clinics or were organized in dedicated telemonitoring groups or teams concerning care centred around the HT application.	5
Value proposition	Better use of resources resulting in reduced time of treatment	Better use of resources was achieved through automation of processes, better means of prioritizing patients based on high quality of data, data overview and sharing, better division of work and interconnection of tasks between healthcare sectors.	6
	Ease of use	Ease of use was established through easy integration to other healthcare IT, recognizable interfaces, and a complete solution for a well-defined issue.	6
	Employee satisfaction	Employee satisfaction was increased when the process of learning how to use the application was considered meaningful. Meaningfulness was experienced through a lower level of distrust between primary and secondary sectors, and the feeling of providing a more interlinked, reflected and better healthcare service. In addition, one respondent highlighted that specialists saw more relevant patients than before using the application.	5
Channels	End-users influencing payers	End-users in the form of patients and healthcare providers were important to influence decision makers and payers when choosing provider company.	5
	Conferences and fora	Conferences and fora were ways of getting in touch with representatives from the political world. However, this was also a channel to healthcare providers. Some companies indicated that healthcare providers were more actively targeted.	5
	Word of mouth	Word of mouth was an important channel to potential customers especially among healthcare providers. This was further illustrated by researchers picking up the telemedicine application if already in use in clinical practice in the department where he/she worked. In addition, some mentioned references from existing HT initiatives as a channel to potential customers.	5
Customers relationships	Personal assistance	Personal assistance was acquired through personal visits by company representatives. This allowed the possibility to educate customers and users in the use of the HT application as well as provide support. Support was established through hotlines in some cases.	6
	Keeping customers	A good connection to the customers through lead users or clinical ambassadors, focusing on accommodating the customers complaints and suggestions was important to keep existing customers. Some also mentioned an easy transition from project phase into clinical practice.	5

(continued)

Table 3. Continued.

BMC element	Theme	Description	Diversity (div)
	Cocreation	Cocreation was a direct effort to include and work with healthcare providers during the process of developing, testing or revising the telemedicine application, including individualized logistics adjustments.	5
Key resources	Technical personnel	IT staff or engineers were responsible for development, maintenance, or technical support. In one case, technical personnel also had the competences to translate clinical issues or problems into technical solutions.	6
	Healthcare personnel	Healthcare personnel took part in the development of the HT application. They differentiated from technical personnel in having responsibility of, and competences to help the actual implementation of the applications through domain knowledge and ability to translate clinical needs to specific solutions.	5
	IT infrastructure	Employee software and hardware, as well as a fast and stabile internet connection, datacentres, and servers.	5
Key activities	Support	Receiving and performing support services for customers and own employees from the starting phase to implementation, and operation.	7
	Development	Activities regarding development, improvement and adjustment to the HT application.	6
	Education	Education of users, lead users and distribution partners.	6
Key partners	Lead users and clinical ambassadors	Healthcare providers were important partners and took part in the development of HT applications.	6
	Local IT-departments	Local IT-departments in the region, hospital ward or municipalities, secured access to local IT-infrastructures.	4
	Other suppliers of healthcare IT	Suppliers of healthcare IT provided integrations to other healthcare IT solutions, mostly to electronic healthcare records.	3
Revenue streams	Licensing	Licenses were either dependent or independent of the number of users. Licenses were highlighted as the primary source of income.	5
	Add-on sales	Sale of coursework, add-on modules to the HT applications, or data extraction and subsequent statistic calculations.	5
	"Free" of charge	Some companies emphasized that paying for their service gave full access to all facets of the application, while others offered maintenance and support without extra costs.	4
Cost structures	IT	IT costs covered hardware, maintenance, telecommunications providers and hosting.	7
	Salaries	The primary cost was indicated to be salaries.	6
	Economy of scale	Costs of scaling was diverse. Several companies had strategically though about scalability while developing the HT application.	5

Revenue streams

Overall, companies generated revenue from *Licensing* and *Add-on sales*. Recurring revenue from licenses was the primary revenue stream for most providers. Additional revenue was made through sales related to coursework, IT add-on modules, and data extraction.

Costs

The top three themes regarding costs were *IT, Salaries, and Economy of scale*. All providers had *IT costs,* ranging from costs related to hardware and maintenance to telecommunication providers and hosting. However, the biggest expense was *Salaries*.

Discussion

This study represents a unique approach to qualitatively analysing the characteristics of the BMs of long-term operational telemedicine providers within a primarily publicly funded healthcare system. In line with previous studies, we identified 22 HT initiatives in operation in Denmark.[21] Companies included for interview operated within various medical specialties, generally had no other products than HT, and were small- and medium-sized enterprises. All except one were privately owned. Similarly to this study, other studies on telemedicine business models do not point to a specific medical specialty related to long-term operational telemedicine.[8,19] Chen et al. found that companies in general provide other products or services in addition to telemedicine solutions.[8] However, we found that most HT companies in Denmark focus solely on telemedicine products and services.

The invisible payer

In a publicly funded healthcare system, it is no surprise that patients and doctors act as end-users and not payers. However, within this type of healthcare system it seems as if companies struggle to identify individuals that have the final say when it comes to making purchases. The dominant solution is a bottom-up approach, but it is not clear if this is a deliberate strategy or a result of being unable to identify and address decision makers. None of the investigated companies offered services directly to patients. In contrast Kho et al. found that some companies use partnerships with insurance companies to reach large numbers of patients, and many companies offered both B2B and B2C services.[18]

Patient experts

Even though all companies try to address value propositions targeting patients, there is little consensus on what the value propositions are. Consequently, no prominent theme related to patient value propositions was identified. An intense focus on healthcare providers seems to be at the expense of patients. Even though patients are the most frequently addressed customer segment, this is not equally reflected in value propositions, channels, and customer relationships. No prominent theme illustrates providers in direct contact with patients to address their needs. Considering that HT often engages patients directly, one would expect a higher degree of patient involvement. Value propositions targeting patients were developed indirectly through healthcare providers acting as 'patient experts'. Other studies highlight the importance of clinician involvement during the design and implementation process, but do not mention direct involvement of patients.[18] This might result in a small group of healthcare providers making decisions on behalf of many patients without their direct involvement. This is incongruent with political goals of increased patient engagement as well as general design concepts for developing value propositions in close communication with the targeted customer segment.[16,17,27]

Costs, patents and evidence

This study supports the current literature by finding value propositions that primarily focus on the needs of healthcare providers. Other studies demonstrate that value creation is generated through accessibility to healthcare, enhanced collaboration between healthcare personnel, and efficiency in the delivery of care through time and cost savings.[6,18] The latter resonates well with value propositions found in this study. However, cost reductions were not explicit, and overall competitive prices on products and cost savings as value propositions were only indicated by three companies. One may argue that reduced time of treatment may reduce costs, but the respondents did not support a causal relation. Our study found an emphasis on human resources. This is typical of knowledge-intensive industries like pharmaceutical companies.[17] In the light of this, it seems surprising that intellectual properties like patents were only mentioned by one of the respondents. Patents are not mentioned by other telemedicine BM studies either.[6,8,19] Valeri et al. argue that sustainable BMs should be dynamic, adapt to new situations, and make use of future potentials of telemedicine.[19] This could pose a paradox, since dynamic and everchanging BMs could have a hard time keeping up with the demand for evidence suggested by evaluation models such as The Model for Assessment of Telemedicine (MAST),[15] since this evidence is especially costly and time consuming within healthcare. Three companies supported a theme of clinical and economic evidence. It seems surprising that respondents did not reference this more

during the interviews, since all except one provided us with documentation on their applications, ranging from reports and evaluations, to peer-reviewed studies. However, the economic evaluation of telemedicine is a recurrent issue.[16]

Data usage

It is noticeable that none of the interviewed companies reported use of user-data or automated analysis as ways of improving value propositions, especially since this is a popular method used by tech companies, and was found by other authors investigating telemedicine business models.[18] Healthcare data jurisdiction could explain why this is not the case, but nonetheless data usage was not highlighted by respondents explicitly. In general, the potential for telemedicine to provide novel patient-related data in big quantities was not emphasised. Telemedicine providers help generate valuable data, potentially benefitting not only patients and healthcare providers, but also researchers, healthcare politics and society at large. Healthcare authorities could use this data to enforce telemedicine applications based on solid arguments rather than assumptions about patients' behaviour, health, and related treatment.

Impacts of personal relations

In general, most providers spent money on technical and healthcare in-house staff salaries. Their main jobs were to either develop value propositions targeting healthcare providers or support customer relations by making personal customer visits. This was the most prominent way of interacting with users throughout all the companies investigated. Findings regarding customer relationships show that personal visits served multiple purposes and had strong impacts throughout all elements of the business models. Thus, in order to achieve a successful telemedicine concept, an issue of paramount importance for companies, end-users, and decision-makers, was establishing strong personal relationships between company representatives and healthcare providers. This has not been explicitly found in other studies, but they do comment on the importance of co-creating value, as well as having end-users help develop value propositions, channels, and improve stakeholder relationships.[6,18] The qualitative nature of this study, providing a deeper understanding of areas such as customer relations, might explain this discrepancy. Even though other studies find that most telemedicine ventures make use of personal assistance,[8] we argue that our study provides novel insights into how these relations are facilitated as well as their impact. These relations seem of higher importance than value propositions based on clinical and economic rationales, and BMs depend on human resources rather than intellectual property rights. However, one may question to what extent a BM built and dependent on personal relations is a scalable business. In addition, we wonder if the full potential of HT can be achieved without direct patient involvement, and a more extensive use of data.

Limitations

Scope is a potential limitation of our study. Firstly, a sole focus on home telemonitoring could complicate comparisons with the results of studies using different subgroups of telemedicine. However, broadening the scope of inclusion could potentially inhibit the ability to find common characteristics within a small sample size. Secondly, determining the success of telemedicine is controversial and could vary widely from the perspective of the provider, healthcare manager or end-user.[14,15] In this study we focused on the transition into clinical practice, hence the decision to include providers solely based on time in operation.

Potentially lost or distorted data is another limitation. Sorting themes by the number of companies supporting each theme had a big influence on the final interpretation. This was necessary to make sense of the high number of themes that resulted from the meaning analysis. It was our impression that consolidating themes further would result in themes that were too heterogeneous and would not incorporate enough detail for themes to be applicable. The large amount of data could have resulted in lost or distorted information, including findings contradicting the central phenomenon highlighted above. Three interviews included more than one respondent, which could have resulted in information bias. However, all companies had the same opportunity to select respondents, and were all briefed in a timely manner to allow them to prepare answers for each question. Other studies provide specific details on costs and revenue streams, and highlight the importance of start-up funding.[8,18] In our study themes regarding costs and revenue streams could benefit significantly from quantitative measures. Key resources, activities, partnerships and costs were placed last in the interview guide. This may have resulted in a hastier and thus less thorough examination of this part of the BMC, and some respondents expressed confidentiality concerns regarding costs and revenue streams. In general, the validity of the results is a concern as all our results are based on the respondents' own opinions and may be biased. To address this, interviews of customer segments like healthcare providers and an evaluation of the evidence could provide a cross reference. Lastly, the findings do not prove a causal relation between BM characteristics and successful integration into clinical practice.

Conclusion

In general value propositions empower healthcare providers to make better use of resources, resulting in reduced time of treatment. Value propositions are delivered through personal relationships with end-users, and revenue is generated via indirect channels to decision makers paying primarily through licensing. Personal relations are essential to all aspects of the business model 1) to secure a strong relationship to lead users and clinical ambassadors; 2) to facilitate work with healthcare providers to develop, test and revise value propositions; 3) to promote user support and education; 4) to establish indirect relations between companies and healthcare managers or decision makers. Healthcare managers and telemedicine providers should be aware of potential pitfalls in current business models. These include lack of identification of decision makers, patient involvement, use of data, and business scalability. In addition, telemedicine companies with the ability to establish strong personal connections could be favoured over companies which provide strong clinical and economic evidence.

Future perspectives

Future studies should analyse the characteristics of failed telemedicine initiatives or BMs from the perspective of the clinic. Studies could investigate telemedicine initiatives other than home telemonitoring, as well as using the findings of this study as a hypothesis for quantitative studies. Lastly, future studies could investigate why personal relations are critical within telemedicine compared to the automated processes used widely in other parts of the tech industry.

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Supplemental Material

Supplemental material for this article is available online.

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